Hospital Treatment Success Rate Evaluation

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1. Aim

To evaluate whether the **new hospital treatment protocol improves patient recovery rates** compared to the existing protocol using the **Two-Proportion Z-Test**. Additionally, perform **correlation analysis** to examine the relationship between patient age and recovery duration, and between treatment cost and recovery success.

2. Problem Statement

A hospital introduced a **new treatment protocol** aimed at improving patient recovery rates. Data were collected from two independent groups:

- Patients treated with the **existing protocol**
- Patients treated with the **new protocol**

The dataset contains the following attributes:

- Total number of patients
- Number of successful recoveries
- Patient age
- Treatment duration (days)
- Treatment cost

The objective is to determine whether the **new protocol significantly improves recovery rates** using a **two-proportion Z-test at a 5% significance level**. Additionally, correlation analysis is performed to study the relationship between **patient age and recovery duration**, and between **treatment cost and recovery success**.

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3. Design Document

The system is designed to analyze hospital treatment effectiveness using statistical hypothesis testing and correlation analysis. The dataset is stored in a CSV file containing patient treatment information for both protocols. Using Pandas for data handling and SciPy for statistical analysis, the program calculates recovery proportions for both treatment groups.

Sample Input

The dataset is stored in a CSV file named hospital_treatment_data.csv and contains patient treatment details.

Protocol,TotalPatients,Success,Age,Duration,Cost

Existing,100,65,45,10,5000

New,100,80,50,8,5500

Existing,120,70,52,11,5200

New,120,90,47,7,5800

Existing,110,68,49,10,5100

New,110,85,46,8,5600

Existing,90,55,54,12,4800

New,90,75,48,9,5300

Existing,95,60,51,11,5000

New,95,78,47,8,5400

Attributes:

- Protocol – Existing or New treatment protocol
- TotalPatients – Total number of patients treated
- Success – Number of successful recoveries
- Age – Average patient age
- Duration – Treatment duration in days
- Cost – Treatment cost

Sample Output

Recovery Rate (Existing Protocol): 0.63

Recovery Rate (New Protocol): 0.81

Z Statistic: -3.12

P Value: 0.0018

Correlation between Age and Recovery Duration: 0.58

Correlation between Treatment Cost and Recovery Success: 0.66

Decision:

Reject Null Hypothesis

Interpretation:

The new treatment protocol significantly improves patient recovery rates compared to the existing protocol.

There is also a moderate positive correlation between patient age and recovery duration, indicating older patients may require longer recovery periods.

Additionally, treatment cost shows a positive relationship with recovery success, suggesting that highe

Tools Used

- Python
- NumPy
- Pandas
- SciPy
- CSV dataset

4. Methodology

Steps involved in the analysis:

1. Collect hospital patient treatment data.
2. Store the dataset in a CSV file.
3. Import the dataset using Python.
4. Separate patients into existing protocol and new protocol groups.
5. Compute recovery proportions for both groups.
6. Apply the **Two-Proportion Z-Test** to test for significant differences.
7. Calculate Z-statistic and p-value.
8. Perform correlation analysis between:

- Patient age and recovery duration
- Treatment cost and recovery success

9. Interpret the statistical results.
10. Draw conclusions for healthcare decision-making.

5. Mathematical Model

Let:

- p_1 = Recovery proportion for existing protocol
- p_2 = Recovery proportion for new protocol

Recovery Proportion

$$p = \frac{\text{Number of Successful Recoveries}}{\text{Total Patients}}$$

Pooled Proportion

$$p = \frac{x_1 + x_2}{n_1 + n_2}$$

Where:

- x_1, x_2 = successful recoveries
- n_1, n_2 = total patients

Z Statistic

$$Z = \frac{p_1 - p_2}{\sqrt{p(1-p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Decision rule:

If **p-value < 0.05**, reject the null hypothesis.

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6. Hypothesis Formulation

Null Hypothesis (H_0)

There is **no significant difference** in recovery rates between the existing and new treatment protocols.

$$H_0 : p_1 = p_2$$

Alternative Hypothesis (H_1)

The **new protocol improves recovery rates** compared to the existing protocol.

$$H_1 : p_2 > p_1$$

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7. Matrix Representation

Example dataset representation:

$$X = \begin{bmatrix} Protocol & TotalPatients & Success & Age & Duration & Cost \end{bmatrix}$$

Example sample:

Protocol	Total	Success	Age	Duration	Cost
Existing	100	65	45	10	5000
New	100	80	50	8	5500

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7. Matrix Representation

Example dataset representation:

$$X = \begin{bmatrix} Protocol & TotalPatients & Success & Age & Duration & Cost \end{bmatrix}$$

Example sample:

Protocol	Total	Success	Age	Duration	Cost
Existing	100	65	45	10	5000
New	100	80	50	8	5500

Flowchart:

Start
Load Patient Dataset
Separate Existing and New Protocol Data
Compute Recovery Proportions
Apply Two-Proportion Z-Test
Calculate Z-statistic and P-value
Perform Correlation Analysis
Interpret Results
Display Conclusion
End

Implementation:

CSV Dataset (hospital_treatment_data.csv)

```
Protocol,TotalPatients,Success,Age,Duration,Cost
Existing,100,65,45,10,5000
New,100,80,50,8,5500
Existing,120,70,52,11,5200
New,120,90,47,7,5800
Existing,110,68,49,10,5100
New,110,85,46,8,5600
```

Python Code

```
import pandas as pd

import numpy as np

from scipy.stats import pearsonr

from statsmodels.stats.proportion import proportions_ztest


# Load dataset

data = pd.read_csv("hospital_treatment_data.csv")


# Separate protocols

existing = data[data['Protocol'] == 'Existing']

new = data[data['Protocol'] == 'New']


# Total successes and patients

success = np.array([existing['Success'].sum(), new['Success'].sum()])
```

```
total = np.array([existing['TotalPatients'].sum(), new['TotalPatients'].sum()])

# Two proportion Z test
z_stat, p_value = proportions_ztest(success, total)

print("Z Statistic:", z_stat)
print("P Value:", p_value)

# Correlation analysis
age_duration_corr, _ = pearsonr(data['Age'], data['Duration'])
cost_success_corr, _ = pearsonr(data['Cost'], data['Success'])

print("\nCorrelation between Age and Recovery Duration:", age_duration_corr)
print("Correlation between Treatment Cost and Recovery Success:",
cost_success_corr)

# Decision
if p_value < 0.05:
    print("\nReject Null Hypothesis")
    print("New protocol significantly improves recovery rates")
else:
    print("\nFail to Reject Null Hypothesis")
```

Output:

```
[Running] python -u "e:\python ds\practice.py"
Z Statistic: -21.076300482398306
P Value: 1.3125067851476866e-98

Correlation between Age and Recovery Duration: 0.6996967194789543
Correlation between Treatment Cost and Recovery Success: 0.97071758286434

Reject Null Hypothesis
New protocol significantly improves recovery rates

[Done] exited with code=0 in 14.374 seconds
```

RUBRICS:

Problem Understanding (10)	Design Document (10)	Implementation (15)	Output (5)	Time Management (5)	Viva (5)	Total (50)

Conclusion:

The **Two-Proportion Z-Test** was applied to evaluate whether the new hospital treatment protocol significantly improves patient recovery rates compared to the existing protocol. Since the calculated p-value is less than the 5% significance level, the null hypothesis is rejected, indicating that the new treatment protocol results in significantly higher recovery rates.